

**Course Code: EET201****Course Name: CIRCUITS AND NETWORKS**

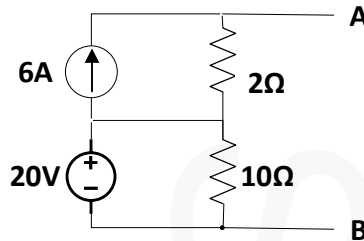
Max. Marks: 100

Duration: 3 Hours

**PART A***Answer all questions. Each question carries 3 marks*

Marks

1. State and explain maximum power transfer theorem in DC circuits. (3)
2. Replace the network given below with a single current source and a resistor. (3)



3. Explain the classification of series RLC circuits based on damping ratio. (3)
  4. Obtain the expression for the voltage across a capacitor discharging through a resistor of resistance R. Assume that the initial voltage of the capacitor is  $V_0$ . (3)
  5. Determine the voltage  $v(t)$  across a  $2\Omega$  resistor, if the current is given by, (3)
- $$I(s) = \frac{2s + 4}{s^2 + 4s + 3}$$
6. Derive the s-domain equivalent circuit of a capacitor having an initial voltage of  $V_0$ . (3)
  7. Explain the phenomenon of neutral shift in three phase 3 - wire systems. (3)
  8. Derive an expression for the Q- factor of series resonant circuits. (3)
  9. Express ABCD parameters in terms of Z parameters. (3)
  10. Determine whether the two port network represented by the following network equations is reciprocal. (3)

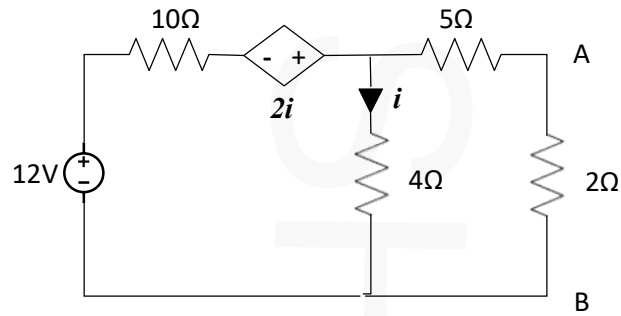
$$V_1 = 3V_2 - 2I_2$$

$$I_1 = 4V_2 - 3I_2$$

**PART B***Answer any one full question from each module. Each question carries 14 marks***Module 1**

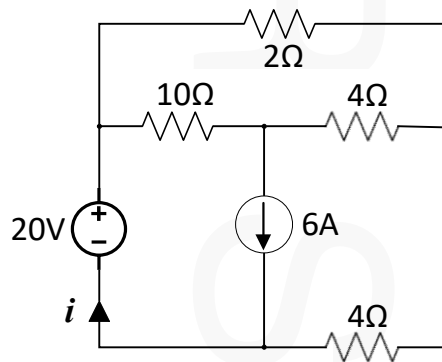
- 11 For the network given below,
  - a) Obtain the Thevenin's equivalent circuit across the terminals A and B. (10)

- b) Determine the power dissipated in the  $2\Omega$  resistance. (4)



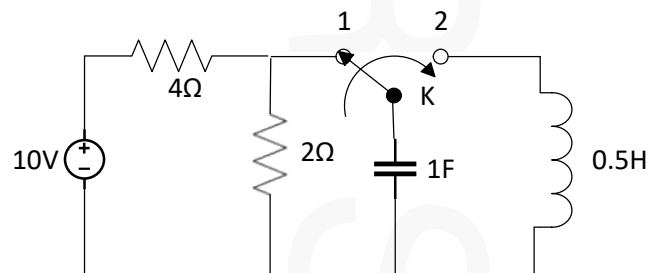
- 12 In the circuit given below,

- a) Find the current  $i$  using superposition theorem. (10)  
 b) Determine the power supplied by the 20V source (4)

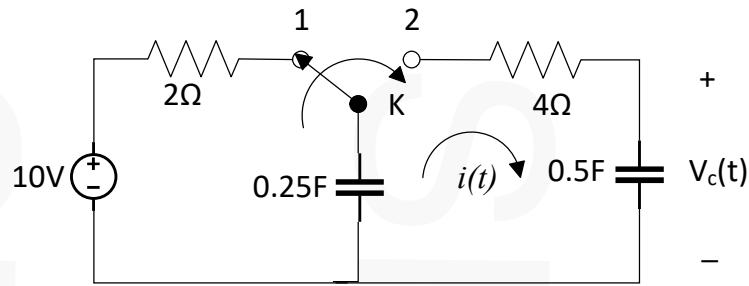


### Module 2

- 13 a) A series RL circuit with  $R = 10\Omega$  is connected to a 50V DC supply at  $t = 0$ . Determine the value of the inductance  $L$  if the current through the inductor attains 50% of its steady state value in 1 seconds. (7)  
 b) The switch  $K$  in the circuit given below has been at position 1 for a long time. At  $t = 0$ , the switch is moved to position 2. Determine the current flowing through the inductor for  $t \geq 0$ . (7)

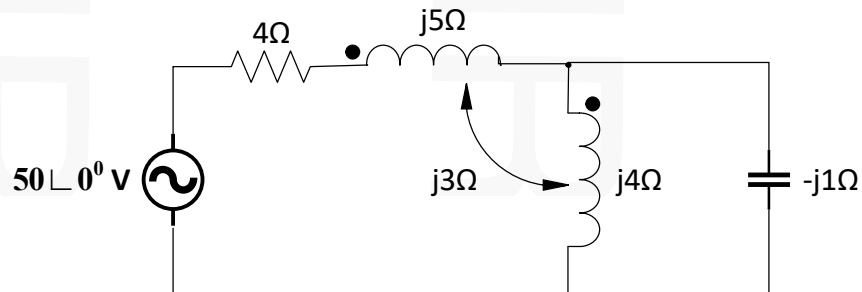


- 14 For the circuit shown below, the switch  $K$ , initially at position 1 for a long time, is changed to position 2 at time  $t = 0$ . Using Laplace transform technique,  
 a) Find the circuit current  $i(t)$  for  $t > 0$ . (8)  
 b) Obtain the expression for the voltage  $V_c(t)$  across the 0.5F capacitor. (6)

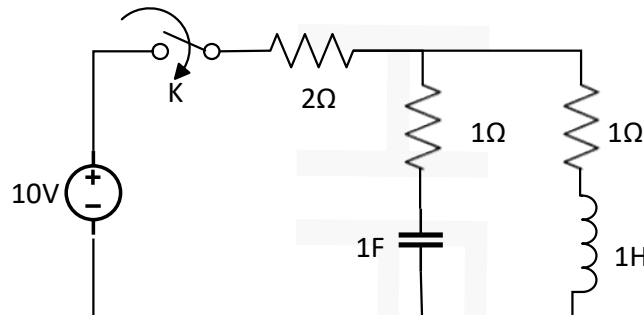


## Module 3

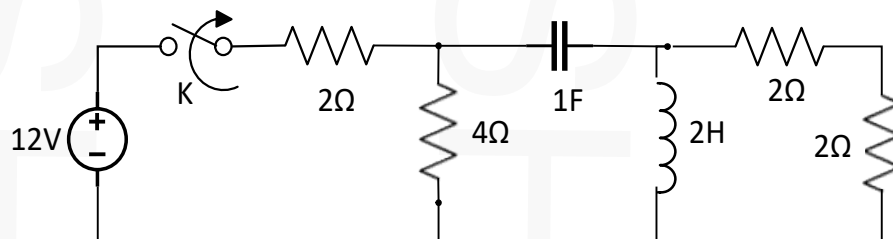
- 15 a) In the circuit given below, find the current flowing through the  $-j1\Omega$  capacitor. (10)



- b) In the circuit given below, the switch K is closed at  $t = 0$ , when the initial current through the inductor is zero and initial voltage on the capacitor is 4 V. Draw the transformed circuit for  $t > 0$  and write the mesh equations in s-domain. (4)



- 16 The switch K in the circuit given below is in closed position for a long time. At  $t = 0$ , the switch is opened.
- a) Determine the transformed circuit for  $t > 0$ . (4)
- b) Find the expression for the voltage across the inductor, for  $t > 0$ , using nodal analysis. (10)

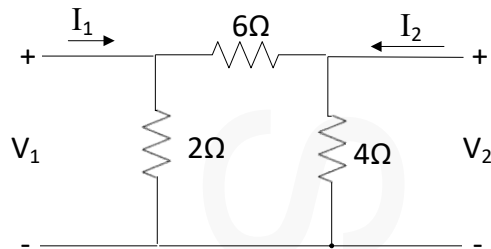


## Module 4

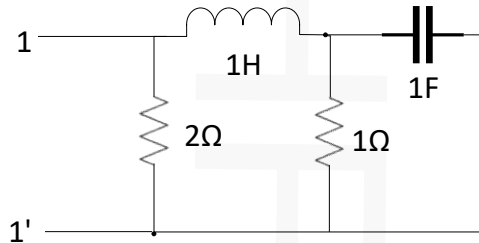
- 17 A resistor, capacitor and an inductor are connected in series with a 230 V, variable frequency AC source. When the supply frequency is varied to 50Hz, a maximum current of 2A flows and the corresponding voltage across the capacitor is 500 V. Determine, (6)
- (i) Resistance, inductance and capacitance of the circuit. (4)
- (ii) Q- factor and bandwidth of the circuit. (4)
- (iii) The source frequencies at which the circuit current is  $\frac{1}{\sqrt{2}}$  times the maximum current.
- 18 A 400 V, three-phase supply feeds an unbalanced three-wire, star-connected load. The branch impedances of the load are  $Z_R = 10\Omega$ ,  $Z_Y = -j5\Omega$  and  $Z_B = j15\Omega$ . Calculate the line currents. (14)

## Module 5

- 19 a) Find the transmission parameters of the network shown in the figure. (8)



- b) Find the driving point impedance of the network given below. (6)



- 20 a) Discuss the series and cascade interconnection of two port networks. (8)
- b) The Y parameters of a two port network are  $Y_{11} = 3\text{S}$ ,  $Y_{12} = -1\text{S}$ ,  $Y_{21} = -1\text{S}$  and  $Y_{22} = 2\text{S}$ . Determine the equivalent T-network. (6)

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